AI Audit Report — HW03 Task 2: Usability Evaluation

Mandatory appendix for every AI-assisted homework (HW#01–HW#06, and Seminar).

1. Student Information

Field	Value
Student name (printed):	DANG TRUONG NGUYEN
Student ID:	23127438
Class / Cohort:	23KTPM3
Assignment ID (e.g., HW#00, HW#02):	HW03 — Task 2 (Usability Evaluation)
Assignment date:	27/07/2026
AI tool(s) used:	Claude Code (Claude Fable 5)
AI tool(s) used:	[X] Yes [ ] No

2. Instructions (read before filling)

- Add one row per AI-generated artifact (test case, script, checklist, OpenAPI spec, JMeter plan, etc.).
- Paste the verbatim prompt — DO NOT paraphrase.
- Paste the verbatim AI output (or include a labelled screenshot in the report).
- Tag the verdict: VALID / INVALID / INCOMPLETE.
- Reasoning must cite a course slide, ISTQB section, or technical RFC.
- Show the corrected artifact with the change highlighted.
- Sample rows are in italic — replace them before submission.

3. Audit Table — one row per artifact

All prompts are verbatim from `plan/PLAN-TASK2.md`; `<<...>>` marks what was pasted in at run time.

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Tool: Claude Code (Claude Fable 5) **Date:** 27/07/2026 **Artifact:** T2·GĐ1 · Research objectives **Prompt (verbatim, Prompt #1):** ROLE: You are a UX researcher helping design a small-sample moderated usability study. CONTEXT — System Under Test: - EShop, a Vietnamese e-commerce demo web app (React + Vite), tested at localhost:5173. - The end-to-end flow under study: Register a new account → search for a product → view detail → add to cart (total ≥ 300,000₫) → checkout applying coupon "SAVE10" → verify the order in Order History. - Study format fixed by the assignment: 7 moderated think-aloud sessions, 1 participant each, SUS (or UEQ-S) after each session, plus	6 candidate objectives O1–O6, each an answerable question tagged with the flow step where evidence appears + the Task 1 bug it is grounded in, plus method notes (O2/O4 overlap; session-risk from BUG-17/BUG-11; n=7 = qualitative signal only).	INCOMPLETE	ISTQB FL – test planning: objectives determine what is observed and how results are interpreted; with n=7 only qualitative questions are answerable. Selecting among candidates is a human planning decision (G9.3 – Analyse), so the output is by design a candidate list.	I selected O1/O2/O6 with written rationale (coverage across three flow segments; O1 answers the mandatory error-recovery probe axis) and rejected O3/O4/O5 with reasons (O4 duplicates the mandatory trust probe; O3 overlaps O2; O5 too diffuse for a 15–25-min session) — documented in `usability-report.md` §1.

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open-ended probe questions. - Known friction candidates from my earlier GUI-checklist testing (Task 1) on these same screens: <<filled by the agent itself from `bug-report.md`: BUG-11/12/16/17/20/33/34/42/44/46>> TASK: Propose 4–6 candidate research objectives for this study. Each objective must be: 1. Observable within a 15–25 minute session on THIS flow (not generic "improve UX"). 2. Phrased as a question the study can actually answer with 7 participants (qualitative signal, not statistical proof). 3. Tagged with where in the flow the evidence would show up (which step/screen). Do NOT write the task scenario or questionnaire yet. Objectives only.				

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Tool: Claude Code (Claude Fable 5) **Date:** 27/07/2026 **Artifact:** T2·GĐ2 · Task scenario **Prompt (verbatim, Prompt #2):** ROLE: You are a UX researcher writing a task scenario for a moderated usability session, in Vietnamese, for Vietnamese participants. CONTEXT: EShop web app. Flow under test: Register → search product → product detail → add to cart (cart total must reach ≥ 300,000₫ for the coupon to apply) → checkout with coupon "SAVE10" → confirm the order appears in order history. Real seeded products the participant can buy: <<filled from `backend/database.js` seed: 5 products, cheapest 4,000,000₫; AirPods Pro 2 = 6,000,000₫>>. TASK: Write ONE	6-sentence goal-oriented scenario (quà sinh nhật, tai nghe chống ồn, ngân sách 6–7 triệu, mã SAVE10 "nhận qua email", "done" theo góc nhìn người dùng) + moderator mapping table phủ đủ 6 bước flow + forbidden-word check. Saved as `task-scenario-draft.md`.	INCOMPLETE	Usability-evaluation lecture / experience-based testing: a scenario that names UI elements leads the participant and invalidates the observation. Grounding in real data was mandatory: the assignment's own example budget style ("under 500,000₫") is infeasible on this catalog (cheapest seeded product = 4,000,000₫).	I re-checked the scenario for leftover UI-instruction words, verified the mapping covers all 6 flow steps, anchored the budget to the real 6,000,000₫ product, and reserved a "post-pilot" section in the file for refinement evidence.

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task scenario (5–8 sentences max) that: 1. Gives the participant a realistic GOAL and motivation (e.g. shopping for an occasion with a budget), NOT instructions. 2. Naturally forces every step of the flow: they have no account yet; they need a specific kind of product; their budget makes the 300,000₫ minimum reachable; they "received a discount code SAVE10 in an email" so using it is part of the goal; and they want to double-check the order was placed. 3. NEVER names any UI element, button label, menu, or screen of the app. Forbidden words include: nút, menu, biểu tượng, giỏ hàng ở góc, trang, tab, click, nhấn vào. 4. States what "done" means from the USER's point of view. Then list, in a separate table				

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(for MY moderator guide only, not read to participants), the mapping: scenario element → flow step it is designed to trigger. OUTPUT both in Vietnamese.				

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Tool: Claude Code (Claude Fable 5) **Date:** 27/07/2026 **Artifact:** T2·GĐ3a · SUS (Vietnamese) **Prompt (verbatim, Prompt #3a):** TASK: Produce the standard 10-item System Usability Scale (SUS) in TWO columns: original English wording and a natural Vietnamese translation suitable for non-technical participants, keeping the strict alternation of positive (odd items) and negative (even items) statements and the 5-point scale ("Hoàn toàn không đồng ý" → "Hoàn toàn đồng ý"). Constraints: 1. Do not paraphrase away the meaning of any item; translate faithfully. 2. Replace "the system" with "trang web mua sắm này". 3. Output as a printable form:	`template/sus-form-vi.md`: (A) EN–VI two-column reference table (polarity +/– marked per item) for translation verification, (B) printable Vietnamese participant form with P1–P7 code + date fields; scoring formula kept off the form.	INCOMPLETE	Course lecture on SUS / measurement validity: SUS scoring assumes strict alternation of positive (odd) and negative (even) items — one mistranslated negative item silently corrupts every score. Translation fidelity must be human-verified.	I reviewed all 10 translations item-by-item against the Brooke originals (hardest: Q8 "cumbersome" → "rườm rà, vướng víu"), confirmed polarity survives translation, and confirmed "the system" → "trang web mua sắm này" throughout.

<table>
<tr><th>(1) Prompt + Tool</th><th>(2) AI Output</th><th>(3) Verdict</th><th>(4) Reasoning (ISTQB)</th><th>(5) Student Fix</th></tr>
<tr><td>instruction sentence for the participant + table with columns | # | Phát biểu | 1 | 2 | 3 | 4 | 5 |, plus fields at the top: Mã người tham gia (P1–P7), Ngày.
Do NOT include the scoring formula on the participant-facing form.</td><td></td><td></td><td></td><td></td></tr>
</table>

(1) Prompt + Tool	(2) AI Output	(3) Verdict	(4) Reasoning (ISTQB)	(5) Student Fix
Tool: Claude Code (Claude Fable 5) **Date:** 27/07/2026 **Artifact:** T2·GĐ3b · Probe questions **Prompt (verbatim, Prompt #3b):** CONTEXT: Same EShop study. My research objectives: <<O1, O2, O6 as selected in Phase 1>>. The assignment requires open-ended probe questions covering AT MINIMUM: clarity, error recovery, speed, and trust. TASK: Write 6–8 open-ended probe questions in Vietnamese to ask AFTER the SUS form, grouped by dimension (clarity / error recovery / speed / trust / + my objectives). Rules: 1. Open-ended and non-leading — no yes/no questions, no question that presumes a problem existed ("Tại sao bạn thấy khó ở bước thanh toán?" is	`template/probe-questions.md`: 7 open-ended Vietnamese questions (C1, C2, E1, E2, S1, T1, T2) grouped by the 4 mandatory dimensions and tied to O1/O2/O6; each with a neutral base + an observation-gated follow-up (↳), plus a suggested asking order.	INCOMPLETE	Non-leading questioning is the interview equivalent of avoiding a leading test oracle: a question that presumes a problem manufactures the data it claims to collect (usability lecture; ISTQB FL – objectivity of test evidence).	I confirmed every base question is answerable without presuming a problem, and enforced the usage rule that ↳ follow-ups fire only on events I personally observed in that session.

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forbidden unless I observed it; give me a neutral base version + a follow-up variant to use only if I DID observe the struggle). 2. Everyday language, no QA/UX jargon. 3. For “trust”, tie to e-commerce reality (dám nhập thông tin, tin đơn đã đặt thành công…).				

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Tool: Claude Code (Claude Fable 5) **Date:** 27/07/2026 **Artifact:** T2·GĐ3c · Session kit **Prompt (verbatim, Prompt #3c):** CONTEXT: Same study. 7 moderated sessions, think-aloud, screen + audio recording with consent. Session budget: <<~25>> minutes (from my dry run). TASK: Produce a session kit in Vietnamese, in Markdown, with exactly these parts: 1. MODERATOR SCRIPT — verbatim opening I read to every participant, covering: we test the PRODUCT not you; think-aloud request with a 1-line demo; you may stop anytime; ask consent to record screen and audio. Keep under 150 words. 2. CONSENT LINE — one sentence the participant agrees to, with	`template/session-kit.md`: verbatim moderator opening (~140 words), consent line, A4 observation template (6 flow steps × 4 columns + probe notes + top-3 footer), 5-phrase neutrality cheat-sheet with the single >2-min intervention rule, plus a pre-session logistics checklist (DB reset, localStorage, recording test).	INCOMPLETE	Moderated-testing protocol (course lecture): identical framing and pre-defined, minimal intervention are what make 7 sessions comparable; an improvised moderator invalidates cross-session comparison.	I filled the placeholders (session budget from my dry run; 2-minute stuck threshold), printed 8 sets (pilot + 7), and read the identical script in every session. Both help interventions (P3, P6) followed the pre-defined rule and were logged verbatim.

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date/name fields. 3. OBSERVATION NOTE TEMPLATE — one A4 page per session: header (P#, date, start/end time), then a row per flow step (Register / Search / Detail / Cart / Coupon+Checkout / Order history) with columns: Hoàn thành? (✓ / trợ giúp / thất bại) | Thời gian | Lỗi & do dự quan sát được | Trích lời think-aloud đáng chú ý. Footer: 3 điều nổi bật. 4. NEUTRALITY CHEAT-SHEET — 5 canned neutral responses, and the single rule for when I MAY intervene (completely stuck > <<2>> minutes).				

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Tool: Claude Code (Claude Fable 5) **Date:** 27/07/2026 **Artifact:** T2·GĐ7 · SUS scoring script **Prompt (verbatim, Prompt #5):** TASK: Write a small self-contained Python script `sus_score.py` that: 1. Reads `sus_responses.csv` with columns: participant,q1,…,q10 (values 1–5). 2. Validates every value is an integer 1–5; on any invalid/missing value, prints which participant+item and exits nonzero (do not silently skip). 3. Computes each participant's SUS score with the standard formula (odd items score-1, even items 5-score, sum × 2.5). 4. Prints a table: per-participant score, then mean, median, min–max, and the adjective/grade band for the mean per the	`result/sus_score.py` + `result/sus_responses.csv` (transcribed from my 7 paper forms). Run output: P1 67.5 · P2 57.5 · P3 30.0 · P4 52.5 · P5 55.0 · P6 35.0 · P7 75.0 — mean 53.2, median 55.0, min–max 30.0–75.0, band "OK".	VALID	ISTQB FL – tool support for testing: deterministic arithmetic belongs in a repeatable script, not in an LLM answer; the script validates inputs and fails loudly instead of silently skipping bad values, making the scoring reproducible for graders.	None required — the script output matched my hand-scored paper forms 7/7. I double-checked the CSV transcription against the paper forms before trusting the run.

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Bangor et al. adjective scale. 5. No external dependencies beyond the standard library.				

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Tool: Claude Code (Claude Fable 5) **Date:** 27/07/2026 **Artifact:** T2·GĐ8 · Findings synthesis **Prompt (verbatim, Prompt #6):** ROLE: You are helping a researcher synthesise moderated usability findings. You must not invent observations. INPUT — my typed raw notes from 7 sessions (P1–P7), one block per session, plus SUS score per participant: <<the 7 `result/session-P*.md` files + SUS table>> TASK: 1. Extract every distinct friction event, keeping its trace: (participant, flow step, what happened, quote if any). If a note is ambiguous, list it under "AMBIGUOUS — researcher must recheck recording"	`findings.md`: 9 themes UF-01→09 (x/7 frequency, bug/design candidate, severity candidate, evidence trace), 3 systemic findings (SYS-01→03), answers to O1/O2/O6, a 3-item AMBIGUOUS list, and the GĐ9 plan (comment on existing Task 1 issues instead of duplicating).	INCOMPLETE	ISTQB FL – defect triage requires evidence-based severity. On recount, the AI's aggregate numbers drifted from the raw notes in 3 places — repeated-clicks 3/7 (actually 4/7), "confused by unreset cart" 6/7 (actually 4/7), "no one used the error message" despite the P1 counter-example recorded in the same output — a plausible-narrative bias where counts bent toward the cleaner story.	I re-verified every count line-by-line against `session-P*.md`, corrected the 3 counts + 4 smaller wordings, resolved the 3 AMBIGUOUS items against the recordings (password errors 7/7; empty-search 4/7 with a count-by-person convention; P6 step 4 = "✓ with error"), and made the final severity decisions (UF-01 Blocker; UF-02 stays Major; UF-06 Major).

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instead of interpreting it. 2. Group similar events into themes (affinity grouping). For each theme report: how many DISTINCT participants hit it (x/7), at which step, and whether evidence suggests a BUG (system misbehaves vs. spec) or a DESIGN ISSUE (system works as built but users stumble). Mark your bug/design suggestion as “candidate — needs researcher confirmation”. 3. Propose a severity for each theme using: Blocker / Major / Minor / Cosmetic — justify each with the participant count and outcome, not adjectives. 4. Explicitly list which themes relate to my research objectives — and answer each objective in 1–2 sentences based ONLY on the data above.				

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Tool: Claude Code (Claude Fable 5) **Date:** 27/07/2026 **Artifact:** T2·GĐ9 · GitHub issue evidence + re-triage **Prompt:** instruction in `findings.md` §5 (add usability evidence as comments on the existing Task 1 issues instead of opening duplicates; propose raising #204 and #240), executed on my request "giúp t phần GĐ9 sử dụng gh" via the GitHub CLI.	7 evidence comments posted on issues #204/#205/#209/#210/#213/#235/#240 (frequency x/7, verbatim think-aloud quotes, observed consequences, links to `findings.md` + session notes; participants referenced by P-codes only); new `usability` label created and applied to all 7; severity re-triage executed on #204 (Major → Blocker) and #240 (Minor → Major) across title/body/labels, preserving the original Task 1 assessment in the body.	INCOMPLETE	ISTQB FL – defect management: new evidence updates the existing defect report (no duplicate reports), and severity re-triage is a human decision based on impact evidence. The AI drafted, formatted, and executed; the re-triage decision came from the human-confirmed findings.	I approved the two severity raises beforehand (decided in `findings.md`), chose to leave #235 at Minor pending my own decision despite the AI's Major suggestion, verified no participant personal data appears in the public comments, and re-checked the final state of both edited issues.

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Tool: Claude Code (Claude Fable 5) **Date:** 27/07/2026 **Artifact:** T2·GĐ10 · Main report assembly **Prompt (conversational):** "rà lại còn chỗ nào liên quan đến AI làm mà tôi chưa update không" + "bổ sung mục main report + pilot đi, tôi chốt rồi" — assemble sections 2–10 of `usability-report.md` from the existing artifacts (scenario, instruments, participants, SUS results, findings, GitHub actions) and write the pilot section per my decision to disclose the missing pilot honestly as a limitation.	Sections 2–11 of `usability-report.md` : scenario & instrument summaries with SUS-choice justification, participants + target profile, preparation & pilot-limitation disclosure, session procedure + intervention log, SUS table (script-scored), severity-ranked findings summary with objective answers, GitHub bug-reporting section, limitations, and a placeholder skeleton for the AI Critique (left for me to write). Also closed out `task-scenario-draft.md` §4 ("no post-pilot changes" decision).	INCOMPLETE	ISTQB FL – test reporting: a test summary report consolidates existing evidence and must not introduce claims beyond it; every number in the report must trace to the underlying artifacts (SUS files, session notes, findings). The pilot disclosure is an integrity decision that only the student can make.	I reviewed each section against its source file (SUS numbers vs `sus_score.py` output, findings table vs `findings.md`, issue links vs GitHub), confirmed the pilot section states the limitation truthfully, and will write the AI Critique (section 11) entirely myself.

4. Summary of AI Accuracy

Aggregate the verdicts from Section 3 and complete the table below.

Metric	Count	Percentage
Total AI-generated artifacts audited	9	100%
VALID (correct, accepted as-is)	1	11%
INVALID (wrong; rejected)	0	0%
INCOMPLETE (acceptable after edits)	8	89%

5. Conclusion — When should AI be used (or not)?

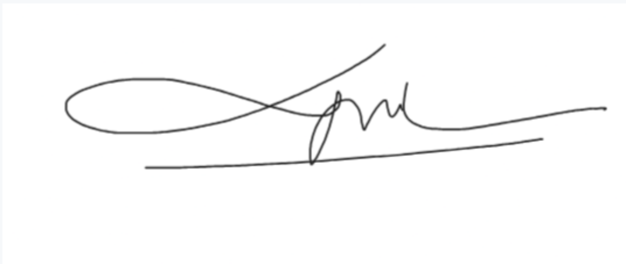
Write 80–150 words describing patterns you observed. Where did AI shine? Where did AI fail? What is your recommendation for using AI in this kind of work in the future?

I used AI at the two ends of the study - designing instruments before the sessions and synthesising after - never during data collection. AI was excellent at building instruments (task scenario, SUS translation, probe questions, session kit) and at deterministic tooling - the SUS scoring script matched my hand-scoring 7/7. But in the synthesis step the AI mis-stated 3 of its own aggregate numbers, each time in the direction that made the narrative cleaner (undercounting an exception, overcounting agreement), and only a line-by-line recount against the raw session notes caught it. My working rule going forward: let AI draft structure and prose, keep every number traceable to raw data, and never accept an aggregate count without re-deriving it. The participant data itself (recruitment, sessions, observation notes, SUS answers) was collected entirely without AI, as the assignment requires.

6. Mandatory Disclosure (paste verbatim)

For Task 2, Claude drafted the research-objective candidates, task scenario, SUS translation, probe questions, session kit, the SUS scoring script, the findings-synthesis draft, and formatted the GitHub issue comments; I selected the objectives, refined and approved every instrument, personally recruited the 7 participants, moderated all sessions, and collected all observation notes and SUS responses without AI — none of the participant list, session notes, or SUS answers is AI-generated. I corrected the AI's synthesis counts against the raw notes before accepting them. The detailed AI Audit Report is attached as Appendix A. I confirm I did not use AI to generate any artifact listed in the prohibited category.

Signature

Student name (printed):	DANG TRUONG NGUYEN
Student ID:	23127438
Class / Cohort:	23KTPM3
Course:	CS423 / CSC13003 – Software Testing
Instructor:	Msc. Tran Thi Bich Hanh
Date:	27/07/2026
Signature:	

References

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